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SUMMARY

Healthcare Data Analysis and Insights is an Excel-based data analytics project focused on analyzing healthcare data to identify meaningful patterns related to patient health profiles, medical history, hospitalization information, and healthcare costs.

PROJECTS

Healthcare Data Analysis and Insights | Excel-based data analytics project analyzing healthcare data to identify patterns related to patient health profiles, medical history, hospitalization information, and healthcare costs.

- Applied data cleaning, data transformation, exploratory analysis, pivot tables, visualization, and dashboard development.
- Cleaned and transformed data across Medical Examinations, Hospitalization Details, and Customer Names tables connected by Customer ID.
- Handled missing values represented by '?' with defined treatment strategies for Month, Year, Smoker, Hospital Tier, City Tier, and State ID.
- Transformed Names into Title, First Name, and Last Name; converted NumberOfMajorSurgeries into numerical data; checked consistency for Heart Issues and Smoker.
- Engineered features including Weight Status from BMI, Diabetes Status from HbA1C, Date of Birth from Year/Month/Date, and Age based on 8 June 2023.
- Integrated data into a consolidated Healthcare sheet using VLOOKUP.
- Analyzed key questions on smoking vs cancer history, transplant history vs surgeries/HbA1C, charges by Weight Status and Diabetes Status, hospital tier/state charge variation, and relationships between age with BMI, HbA1C, and charges.
- Built visualizations including pie/donut charts, column/bar charts, and scatter plots.
- Designed an interactive Healthcare Dashboard with slicers for Weight Status and Diabetes Status.
- Demonstrated end-to-end workflow: Raw Data → Data Cleaning → Data Transformation → Feature Engineering → Data Integration → EDA → Visualization → Dashboard → Business Insights.

TECHNICAL SKILLS

Microsoft Excel
Excel Pivot Tables
Excel VLOOKUP
Excel Charts
Excel Slicers
Data Cleaning
Data Transformation
Data Integration
Feature Engineering
Exploratory Data Analysis
Pivot Table Analysis
Data Visualization
Dashboard Development
Healthcare Data Analysis
Problem Solving
Data Interpretation
Pattern Identification
Business Question Development

Insight Generation
Data-Driven Decision Making
SQL
Power BI
Python
Business Analytics

BUSINESS OBJECTIVES

Clean and prepare the healthcare dataset for analysis; identify and handle missing values; standardize inconsistent healthcare-related fields; transform raw fields into meaningful analytical categories; create patient-level health indicators; analyze healthcare charges across different patient groups; understand relationships between age, BMI, HbA1C, and healthcare charges; compare healthcare charges across hospital tiers and states; analyze health conditions based on smoking and transplant history; build an interactive healthcare dashboard for decision-making.

BUSINESS RECOMMENDATIONS

Monitor healthcare cost differences across patient health categories; identify patient groups associated with higher healthcare charges; compare healthcare costs across hospital tiers; investigate geographic differences in healthcare expenditure; monitor relationships between age and important health metrics; use patient health indicators to support resource planning; use interactive filtering to investigate specific patient segments.

BUSINESS IMPACT

Better understanding of healthcare cost patterns; improved visibility into patient health profiles; identification of important health-related trends; improved comparison of hospital and geographic costs; more effective healthcare resource planning; faster access to meaningful healthcare information; improved data-driven decision-making.

CHALLENGES FACED

Handling missing values represented by '?'; converting inconsistent data into usable formats; cleaning non-numeric surgery values; splitting customer names correctly; creating meaningful BMI categories; creating diabetes categories from HbA1C; combining multiple date fields; calculating customer age; integrating multiple tables using Customer ID; selecting appropriate visualizations for each analytical question; designing an interactive dashboard that remains easy to interpret.

LIMITATIONS

Analysis is dependent on the quality of the provided dataset; missing or inconsistent values may affect analytical accuracy; healthcare charges alone may not explain the complete cost structure; correlation analysis does not establish causation; dataset may not represent the entire healthcare population; analysis is based on available fields and historical dataset information.

FUTURE SCOPE

Automating the data-cleaning process; adding more advanced statistical analysis; performing predictive healthcare-cost analysis; developing patient-risk segmentation; adding time-based healthcare trend analysis; building the solution in Power BI or Tableau; adding automated KPI monitoring; developing machine-learning models for healthcare cost prediction; creating more advanced geographic analysis.

PROJECT STRUCTURE

Healthcare-Data-Analysis/; Dataset/Healthcare Dataset.xlsx; Dashboard/Healthcare Dashboard.xlsx; Documentation/Healthcare Analysis Report.pdf; Screenshots/Healthcare Dashboard.png; README.md

FINAL PROJECT STATEMENT

Healthcare Data Analysis and Insights demonstrates how raw healthcare data can be cleaned, transformed, analyzed, and visualized to uncover meaningful patterns in patient health profiles and healthcare costs. The project combines Excel data preparation, analytical techniques, visualization, and interactive dashboard design to create a practical healthcare analytics solution for data-driven decision-making.